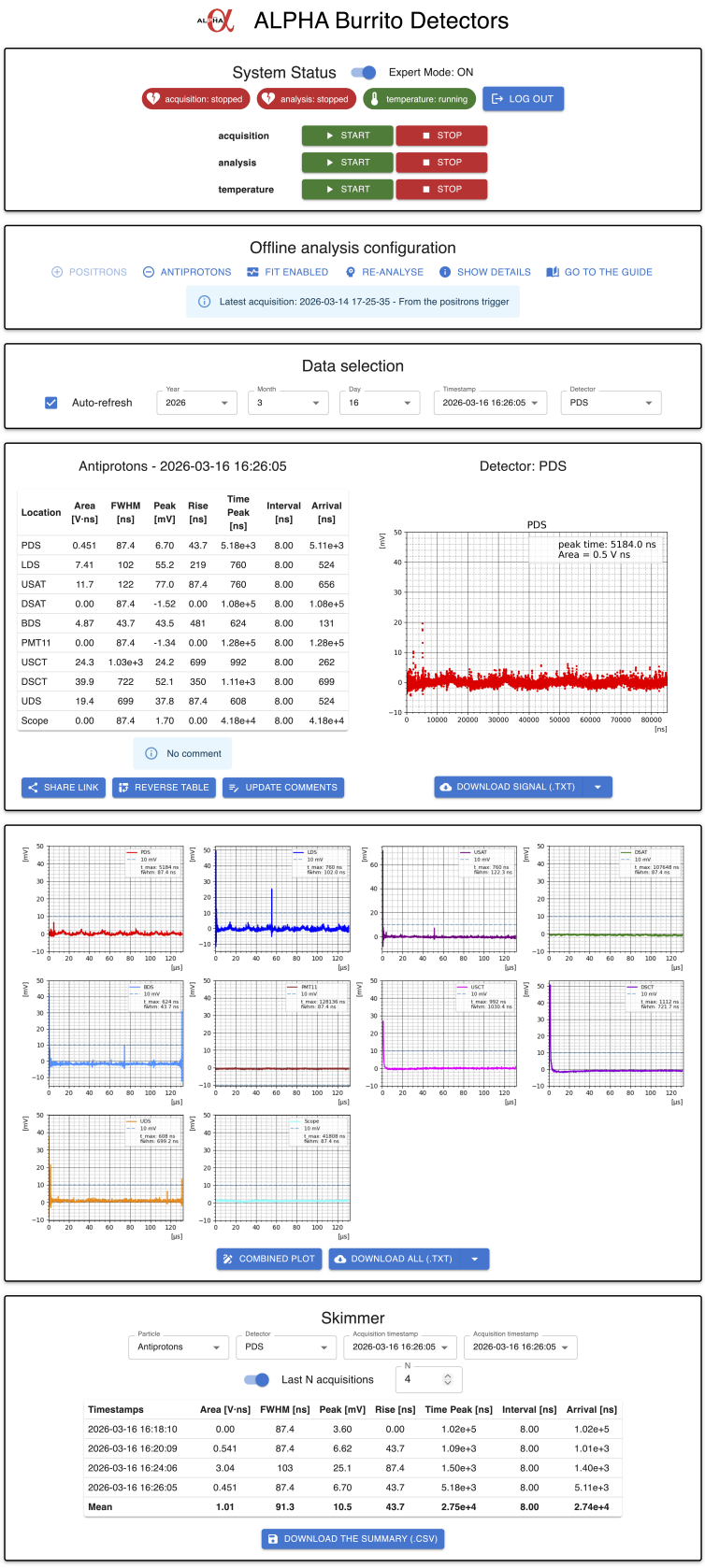
Alpha Burritos

Alpha Burritos is a modern, interactive React application developed for CERN's ALPHA experiment. It provides researchers and analysts with a user-friendly interface to browse, visualize, and analyze signal data from the experiment's "burrito detectors"—specialized detection systems that capture antimatter annihilation events.

The application communicates with a custom RESTful API to fetch and display critical data, including JSON parameter files and detector-specific PNG images captured at various timestamps. By consolidating multiple data streams into a single, responsive interface, Alpha Burritos simplifies the complex task of antimatter detection analysis and monitoring.



Features

- **Temperature Monitoring:** View real-time temperature readings from multiple detector hosts (pitaya boards) in a compact, color-coded table. The temperature data is fetched from the backend and auto-refreshes every 10 seconds. The table uses a green background to match the application's button style.

- **Year/Month/Day Selection:** Choose the year, month, and day to filter available data files.
- **Timestamp Selection:** Select a specific acquisition timestamp from available JSON files.
- **Parameter Display:** View key parameters (area, FWHM, peak, rise, time peak, dt, arrival) for each detector.
- **Combined Image View:** See the combined detector image for the selected acquisition.
- **Detector Selection:** Choose a specific detector (PDS, BDS, DSAT, USAT, PMT11, etc.) to view its processed image.
- **Signal Download:** Download the raw signal as a text file for the selected detector and timestamp.
- **Download All Signals:** Download signal files for all available detectors at once.
- **Comments:** Add and edit comments for specific acquisition files.
- **Auto-Refresh:** Automatically refresh the list of available files.
- **Skimmer:** Browse and export a range of acquisitions and parameters as CSV.
- **Responsive UI:** Built with React and styled for clarity and usability.
- **Authentication:** Secure login for authorized users.
- **Choose Configuration:** Select the analysis mode ("positrons" or "antiprotons") and toggle the fit option for the current session.
- **Re-analyse:** Trigger a new analysis of the currently selected file with the current configuration.

Project Structure

```
src/
  main.jsx                        # React entry point
  main.css                        # Global styles
  app/
    App.jsx                       # Main React component
    LoginForm.jsx                 # User authentication form
    MainTitle.jsx                 # App title and logo
    MessageAlert.jsx              # Reusable alert/message component
  assets/
    ALPHA_Logo_png.png            # ALPHA experiment logo
  configuration/
    ChooseConfiguration.jsx       # Select analysis mode
(positrons/antiprotons) and fit option
    ConfigTable.jsx               # Displays configuration details in a
table
  dataDisplay/
    DataDisplay.jsx               # Container for detector data
visualization
    DetectorImage.jsx             # Shows image for a selected detector
    DetectorImageAll.jsx          # Shows combined detector image (all
detectors)
    DownloadButton.jsx            # Button to download the signal as a
text file
    DownloadAllButton.jsx         # Button to download all signals for
all detectors
    Parameters.jsx                # Displays detector parameters from
JSON
    Skimmer.jsx                   # Browse and export a range of
acquisitions as CSV
```

```
  dataSelection/
    AutoRefresh.jsx               # Toggle auto-refresh of file list
    DataSelection.jsx             # Container for data selection controls
    DateSelector.jsx              # Dropdowns for year/month/day
selection
    DetectorSelector.jsx          # Dropdown for detector selection
    TimeStampSelector.jsx         # Dropdown for timestamp/file selection
  systemStatus/
    SystemStatus.jsx              # System status panel with expert-mode
controls
    TemperatureDisplay.jsx        # Real-time temperature readings for
detector hosts
    WorkerMonitor.jsx             # Monitors Python worker process status
```

How It Works

1. **Select Year/Month/Day:**
   Use the dropdowns to pick a year, month, and day. The app fetches available JSON files for that period.

2. **Select Timestamp:**
   Choose a file (timestamp) from the dropdown. The app loads its parameters and images.

3. **View Parameters:**
   The `Parameters` panel displays key values for each detector.

4. **View Images:**
   - An image containing subplots (one signal per subplot) is shown by default.
   - Use the "Combined plot" button to switch to a single plot with all signals overlaid.
   - Use the detector dropdown to view images for individual detectors.

5. **Download Signals:**
   - Use the "Download the signal" button to download the raw signal for the selected detector.
   - Use the "Download the signals" button to download signals for all available detectors at once.

6. **Add Comments:**
   Add notes or observations about specific acquisition files for collaborative research.

7. **Auto-Refresh:**
   Enable auto-refresh to automatically update the list of available files every 2 seconds.

8. **Skimmer:**
   Use the Skimmer tool to select a range of acquisitions, view their parameters in tabular form, and export as CSV.

9. **Authentication:**
   Login is required to access the main features. Credentials are checked via the backend API.

10. **Choose Configuration:**
    Use the "Choose configuration" panel to select the analysis mode ("positrons" or "antiprotons") and toggle the fit option for the current session.

11. **Re-analyse:**
    Use the "Re-analyse" button to trigger a new analysis of the currently selected file with the current configuration.

API Endpoints

The React app expects the following backend API (see `server.js`):

- **Test if the API is working:**
  `GET /api/test`
  - Returns a JSON file to confirm that the API is working.
  - Example: `/api/test`
  - Response:

```
{ "message": "API is working" }
```

- **List JSON Files:**
  `GET /api/:year/:month/:day/json`
  - Returns a list of JSON filenames for the specified year, month, and day.
  - Example: `/api/2025/06/02/json`
  - Response:

```
["data-2025-06-02_08-25-03.json", "file-2025-06-02.json"]
```

- **Get JSON File Content:**
  `GET /api/json/:jsonFilename`
  - Returns the parsed JSON content of the specified file.
  - Example: `/api/json/data-2025-06-02_08-25-03.json`
  - Response:

```
{ "PDS": { "area": ..., ... }, ... }
```

- **Get Combined Image (single plot, all signals):**
  `GET /api/Together/:imageName`
  - Returns the PNG image with all signals overlaid in a single plot (from the `Together` subdirectory).

  - Example: `/api/Together/data-2025-06-02_08-25-03.png`
  - Response:
    Binary PNG image.

- **Get Subplots Image (default):**
  `GET /api/Same/:imageName`
  - Returns the PNG image containing subplots (one signal per subplot) for all signals (from the `Same` subdirectory).
  - Example: `/api/Same/data-2025-06-02_08-25-03.png`
  - Response:
    Binary PNG image.

- **Get Detector-Specific Image:**
  `GET /api/img/:detector/:imageName`
  - Returns the PNG image for a specific detector.
  - Example: `/api/img/PDS/data-2025-06-02_08-25-03.png`
  - Response:
    Binary PNG image.

- **Get Detector-Specific Signal:**
  `GET /api/signal/:detector/:jsonFilename`
  - Returns the signal as a text file for a specific detector and JSON file.
  - Example: `/api/signal/PDS/data-2025-06-02_08-25-03.json`
  - Response:
    Text file

- **Get Detector-Specific Signal (CSV):**
  `GET /api/signal/csv/:detector/:jsonFilename`
  - Returns the signal as a CSV file for a specific detector and JSON file.
  - Example: `/api/signal/csv/PDS/data-2025-06-02_08-25-03.json`
  - Response:
    CSV file

- **Get Comments:**
  `GET /api/comments/:jsonFilename`
  - Retrieves any saved comments for a specific JSON file.
  - Example: `/api/comments/data-2025-06-02_08-25-03.json`
  - Response:

    ```
    { "comment": "This detection shows interesting peak values." }
    ```

  - Returns `{ "comment": null }` if no comment exists.

- **Post Comments:**
  `POST /api/comments/:jsonFilename`
  - Saves a comment for a specific JSON file.
  - Example: `/api/comments/data-2025-06-02_08-25-03.json`
  - Request Body:

```
{ "comment": "This detection shows interesting peak values." }
```

  - Response:

```
{ "success": true, "message": "Comment updated successfully" }
```

- **Get/Update Configuration:**
  - `GET /api/configuration` — Get the current configuration (from `ANALYSIS_DIR/configurations/configuration.json`).
  - `POST /api/configuration` — Update the configuration (expects JSON body).
- **Get the Latest Dump Timestamp:**
  `GET /api/latest`
  - Returns the latest dump timestamp.
  - Example: `/api/latest`
  - Response:

```
{ "latest": "2025-08-25_11-04-04" }
```

- **Re-analyse a JSON File:**
  `GET /api/reanalyse/:filename`
  - Triggers a re-analysis for the specified JSON file.
- **Check Python Worker Status:**
  `GET /api/status/:pidfile`
  - Checks if a Python worker process is running, based on the given pidfile name.
  - Example: `/api/status/worker`
  - Response:

```
{ "worker": "running" }
```

or

```
{ "worker": "stopped" }
```

- **Authentication:**
  - `POST /api/login` — Authenticate user (expects `{ login, password }` in body).
  - `GET /api/profile` — Returns user info if authenticated.
  - `POST /api/logout` — Logout user and clear authentication cookie.
- **Start a Python Worker Script (requires authentication):**
  `POST /api/start/:name`
  - Starts a Python worker script in the background. Allowed names: `acquisition`, `analysis`, `temperature`.
  - Example: `/api/start/acquisition`
  - Response:

```
{ "success": true, "script": "START_ACQUISITION.py", "pid": 12345
}
```

  - Returns `409` with `{ "error": "acquisition is already running" }` if already running.
- **Stop a Python Worker Script (requires authentication):**
  `POST /api/stop/:name`
  - Stops a running Python worker script. Allowed names: `acquisition`, `analysis`, `temperature`.
  - Example: `/api/stop/analysis`
  - Response:

```
{ "success": true, "script": "ONLINE_ANALYSIS.py", "pid": 12345 }
```

  - Returns `{ "success": true, "message": "analysis is not running" }` if not running.
- **Get Temperature Data:**
  `GET /api/temperature`
  - Returns the latest temperature readings for all detector hosts.
  - Example: `/api/temperature`
  - Response:

```
{
  "hosts": {
    "rp-f0a821.local": { "temp_c": 41.59, "error": null },
```

```
      "rp-f073bf.local": { "temp_c": 42.82, "error": null }
    },
    "timestamp_utc": "2025-10-08 08:42:16"
  }
```

Error Handling

- Returns `404` if a file or image is not found.
- Returns `400` for invalid requests.
- Returns `500` for server errors.

Directory Structure

The API expects files to be organized as:

```
MAIN_DIR/
  YYYY/
    MM/
      DD/
        JSON/
          data-YYYY-MM-DD.json
          comments.json
        Together/
          data-YYYY-MM-DD.png
        Same/
          data-YYYY-MM-DD.png
        PDS/
          data-YYYY-MM-DD.png
          data-YYYY-MM-DD.txt
        BDS/
          data-YYYY-MM-DD.png
          data-YYYY-MM-DD.txt
        ...
```

Development

Prerequisites

- Node.js (v18+ recommended)
- npm

Install dependencies

```
npm install
```

Start the React App

Development mode:

```
npm run dev
```

The app will be available at http://localhost:5173 by default.

For production deployments, it is recommended to use a web server such as **NGINX** to serve the built frontend.
See the NGINX documentation for setup instructions and configuration examples.

Start the Backend API

You have two options:

Option 1: Run directly with Node.js (recommended for development):

```
node server.js
```

Option 2: Run with PM2 (recommended for production):

```
# Start the server
pm2 start server.js --name burritos

# Save the process list
pm2 save

# Set up PM2 to start on system boot
pm2 startup

# Follow the instructions and run the command output by the previous step
```

- View logs: `pm2 logs burritos`
- Stop the server: `pm2 stop burritos`
- Restart the server: `pm2 restart burritos`

The API will be available at http://localhost:3001 by default.

Customization

- **API URL:**
  If your backend runs on a different host or port, update the API URLs in the React components (e.g.,

`Parameters.jsx`, `DetectorImage.jsx`, etc.).

- **Data Directory and Environment Variables:**
  The backend expects data in a specific directory structure (see API documentation above).
  The main directory of the data and other sensitive configuration must be declared in a `.env` file at the project root.

  Example `.env` file:

  ```
  MAIN_DIR=/absolute/path/to/your/data
  ANALYSIS_DIR=/absolute/path/to/your/analysis
  PORT=3001
  USER_LOGIN=your_login
  USER_PASSWORD_HASH=your_bcrypt_hash
  JWT_SECRET=your_jwt_secret
  PYTHON_PATH=/usr/bin/python3
  ```

  - `MAIN_DIR`: Absolute path to the main data directory (required).
  - `ANALYSIS_DIR`: Absolute path to the analysis directory (required).
  - `PORT`: Port for the backend server (default: 3001).
  - `USER_LOGIN`: Username for authentication (required).
  - `USER_PASSWORD_HASH`: Bcrypt hash of the password for authentication (required).
  - `JWT_SECRET`: Secret key for JWT authentication (required).
  - `PYTHON_PATH`: Path to the Python executable used for analysis scripts. If not specified, defaults to `python3`.

  Note: Never commit your `.env` file to version control as it may contain sensitive information.

Useful Resources

- React Documentation
- Vite Documentation
- Express Documentation
- Node.js Documentation
- PM2 documentation
- NGINX Documentation
- JavaScript Reference (MDN)
- JSON Format
- RESTful API Design
- GitHub Guides
- ALPHA Experiment (CERN)

This project uses React, Vite, and Express. For backend API details, see `server.js`.

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